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Philip Mocz

Employment

2025–present **Two Sigma**, Quantitative Software Engineer.
2024–2025 **Flatiron Institute, Center for Computational Astrophysics**, Software Engineer.
2021–2024 **Lawrence Livermore National Laboratory**, Computational Physicist.
2020–2021 **Princeton University**, Lyman Spitzer Jr. Postdoctoral Research Fellow.
2017–2020 **Princeton University**, NASA Einstein Postdoctoral Research Fellow.

Education

2012–2017 **Harvard University**, *Ph.D. Astrophysics.*
Secondary Field in Computational Science and Engineering (2015)
Title: *Moving mesh magnetohydrodynamics: magnetic processes in star formation and cosmology*
Advisor: Lars Hernquist

2008–2012 **Harvard University**, *A.B. Mathematics and Astrophysics.*
Summa cum laude w/highest honors, Phi Beta Kappa

Research Interests

Theoretical Astrophysics and Cosmology. Dark Matter, Cosmological Structure Formation, Stellar Astrophysics, Star Formation, Galaxy Evolution/Feedback, Black Hole Physics
Computational Physics. Computational Fluid Dynamics, Magnetohydrodynamics, Turbulence, Differentiable Simulations, Numerical Methods, High-Performance Computing

Awards and Fellowships

2023 Directors Science & Technology Award: Next-gen Simulation Technologies for the Exascale Era
2020 Buchalter Cosmology Prize
2017–2020 NASA Einstein Fellowship
2017 Eric Keto Prize for best thesis in theoretical astrophysics (Harvard)
2016 Harvard Merit Fellowship
2015–2017 NASA Earth and Space Science Fellowship (NESSF)
2012–2015 NSF Graduate Research Fellowship
2012 Peirce Fellowship (Harvard)
2012 Derek Bok Center Certificate of Distinction in Teaching (Harvard)
2008–2012 John Harvard Scholar
2011 Phi Beta Kappa (Harvard)
2011 Leo Goldberg Prize for Astronomy Junior Thesis (Harvard)
2011 CAS Vacation Scholarship (Swinburne Univ. of Technology)
2010 Weissman International Internship Program Scholarship
2009 Detur Prize (Harvard)
2009 Harvard College Program for Research in Science and Engineering Fellow

2008 Intel Science Talent Search Scholarship

2006 AAS Priscilla and Bart Bok Award

Refereed Publications

59. **Custom Colors: A Module for Computing Synthetic Photometry On-the-Fly in Stellar Evolution Calculations, Integrated with MESA and Usable with Other Stellar Evolution Codes**
Miller, N.; Joyce, M.; Mocz, P.; 2026, submitted

58. **Variability in supermassive black-hole accretion rates in fuzzy dark matter cores due to black-hole wandering**
Ludwig, E.; Mocz, P.; Robles, V.H.; 2026 Phys. Rev. D, 114, 043070

57. **Jaxion: A JAX package for Fuzzy Dark Matter**
Mocz, P.; 2026 Journal of Open Source Software, 11, 9542

56. **Strong Stellar Diffusion from Wave DM Cosmological Simulation and Potential Unified Origin for dSphs, UFGs, and UDGs**
Pozo, A.; Broadhurst, T.; Zhang, J.; Oguri, M.; Umetsu, K.; Emami, R.; Hernquist, L.; Mocz, P.; Vogelsberger, M.; 2026, submitted

55. **Self-consistent Nonlinear Classical Cepheid Pulsations during Stellar Evolution with MESA**
Farag, E.; Bellinger, E.P.; Mocz, P.; Kalici, S.; Smolec, R.; Kanbur, S.; Bettwy, K.; Lindsay, C.; 2026 ApJ, 1002, 172

54. **A smooth filament origin for distant prolate galaxies seen by JWST and HST**
Pozo, A.; Broadhurst, T.; Emami, R.; Mocz, P.; Vogelsberger, M.; Hernquist, L.; Conselice, C.J.; Luu, H.N.; Smoot, G.F.; Windhorst, R.; 2026 Nature Astronomy, 10, 306

53. **Infalling ultra-faint dwarfs as emissaries of the Axiverse**
Pozo, A.; Broadhurst, T.; Luu, H.N.; Smoot, G.; Umetsu, K.; Chiueh, T.; Schive, H.-Y.; Emami, R.; Hernquist, L.; Mocz, P.; Vogelsberger, M.; 2025 Astronomy & Astrophysics, 704, A34

52. **Development of Turbulence in Postshock Regions**
Foley, M.M.; Mocz, P.; Burkhart, B.; Hernquist, L.; Goodman, A.; 2025 ApJ, 992, 164

51. **Validation of FLASH for magnetically driven inertial confinement fusion target design**
Ellison, C.L.; ...; Mocz, P.; ...; 2025 Physics of Plasmas, 32, 093902

50. **Galaxy formation with wave/fuzzy dark matter: The core-halo structure and the solitonic imprint**
Pozo, A.; Emami, R.; Mocz, P.; Broadhurst, T.; Hernquist, L.; Vogelsberger, M.; Smith, R.; Tremblay, G.; Narayan, R.; Steiner, J.; Grindlay, J.; Smoot, G.; 2025 Astronomy & Astrophysics, 699, A308

49. **Diverse dark matter haloes in two-field fuzzy dark matter**
Luu, H.N.; Mocz, P.; Vogelsberger, M.; Pozo, A.; Broadhurst, T.; Tye, S.-H.H.; Liu, T.; Fung, L.W.H.; Smoot, G.F.; Emami, R.; Hernquist, L.; 2025 Phys. Rev. D, 111, L121302

48. **Machine learning visualization tool for exploring parameterized hydrodynamics**
Jekel, C.F.; Sterbentz, D.M.; Stitt, T.M.; Mocz, P.; Rieben, R.N.; White, D.A.; Belof, J.L.; 2024 Machine Learning: Science and Technology, 5, 045048

47. **Tracing 3-D Magnetic Field Structure Using Dust Polarization and the Zeeman Effect**
Shane, B.; Burkhart, B.; Fissel, L.; Clark, S.E.; Mocz, P.; Foley, M.M.; 2024, submitted

46. **An attractive model: simulating fuzzy dark matter with attractive self-interactions**
Painter, C.A.; Boylan-Kolchin, M.; Mocz, P.; Vogelsberger, M.; 2024 MNRAS, 533, 2454

45. **Performance portable graphics processing unit acceleration of a high-order finite element multiphysics application**

Stitt, T.; Belcher, K.; Campos, A.; Kolev, T.; Mocz, P.; Rieben, R.; Skinner, A.; Tomov, V.; Vargas, A.; Weiss, K.; 2024 Journal of Fluids Engineering, 146, 041102

44. **Dwarf galaxies united by dark bosons**
Pozo, A.; Broadhurst, T.; Smoot, G.F.; Chiueh, T.; Luu, H.N.; Vogelsberger, M.; Mocz, P.; 2024 Phys. Rev. D, 109, 083532

43. **Nested solitons in two-field fuzzy dark matter**
Luu, H.N.; Mocz, P.; Vogelsberger, M.; May, S.; Borrow, J.; Tye, S.-H.H.; Broadhurst, T.; 2024 MNRAS, 527, 4162

42. **Cosmic web dissection in fuzzy dark matter cosmologies**
Dome, T.; Fialkov, A.; Sartorio, N.; Mocz, P.; 2023 MNRAS, 525, 348

41. **Growth or decay - I: universality of the turbulent dynamo saturation**
Beattie, J.R.; Federrath, C.; Kriel, N.; Mocz, P.; Seta, A.; 2023 MNRAS, 524, 3201

40. **Structure, Kinematics, and Observability of the Large Magellanic Cloud's Dynamical Friction Wake in Cold versus Fuzzy Dark Matter**
Foote, H.R.; Besla, G.; Mocz, P.; Garavito-Camargo, N.; Lancaster, L.; Sparre, M.; Cunningham, E.C.; Vogelsberger, M.; Gómez, F.A.; Laporte, C.F.P.; 2023 ApJ, 954, 163

39. **Cosmological Structure Formation and Soliton Phase Transition in Fuzzy Dark Matter with Axion Self-Interactions**
Mocz, P.; Fialkov, A.; Vogelsberger, M.; Boylan-Kolchin, M.; Chavanis, P.-H.; Amin, M.A.; Bose, S.; Dome, T.; Hernquist, L.; Lancaster, L.; Notis, M.; Painter, C.; Robles, V.H.; Zavala, J.; 2023 MNRAS, 521, 2608

38. **On the cosmic web elongation in fuzzy dark matter cosmologies: Effects on density profiles, shapes, and alignments of haloes**
Dome, T.; Fialkov, A.; Mocz, P.; Schäfer, B.M.; Boylan-Kolchin, M.; Vogelsberger, M.; 2023 MNRAS, 519, 4183

37. **Small-scale structure in vector dark matter**
Amin, M.A.; Jain, M.; Karur, R.; Mocz, P.; 2022 JCAP, 08, 014

36. **The density distribution and physical origins of intermittency in supersonic, highly magnetised turbulence with diverse modes of driving**
Beattie, J.R.; Mocz, P.; Federrath, C.; Klessen, R.S.; 2022 MNRAS, 517, 5003

35. **Energy balance and Alfvén Mach numbers in compressible magnetohydrodynamic turbulence with a large-scale magnetic field**
Beattie, J.R.; Krumholz, M.R.; Skalidis, R.; Federrath, C.; Seta, A.; Crocker, R.M.; Mocz, P.; Kriel, N.; 2022 MNRAS, 515, 5267

34. **A multi-shock model for the density variance of anisotropic, highly-magnetised, supersonic turbulence**
Beattie, J.R.; Mocz, P.; Federrath, C.; Klessen, R.S.; 2021 MNRAS, 504, 4354

33. **Toward Cosmological Simulations of Dark Matter on Quantum Computers**
Mocz, P.; Szasz, A.; 2021 ApJ, 910, 29

32. **The Catalogue for Astrophysical Turbulence Simulations (CATS)**
Burkhart, B.; Appel, S.M.; Bialy, S.; Cho, J.; Christensen, A.J.; Collins, D.; Federrath, C.; Fielding, D.B.; Finkbeiner, D.; Hill, A.S.; Ibáñez-Mejía, J.C.; Krumholz, M.R.; Lazarian, A.; Li, M.; Mocz, P.; Mac Low, M.-M.; Naiman, J.; Portillo, S.K.N.; Shane, B.; Slepian, Z.; Yuan, Y.; 2020 ApJ, 905, 14

31. **Galaxy Formation with BECDM - II. Cosmic Filaments and First Galaxies**
Mocz, P.; Fialkov, A.; Vogelsberger, M.; Becerra, F.; Shen, X.; Robles, V.H.; Amin, M.A.; Zavala, J.; Boylan-Kolchin, M.; Bose, S.; Marinacci, F.; Chavanis, P.-H.; Lancaster, L.; Hernquist, L.; 2020 MNRAS, 494, 2027

30. **Fuzzy Dark Matter Soliton Cores around Supermassive Black Holes**
Davies, E.Y.; Mocz, P.; 2020 MNRAS, 492, 5721

29. **Dynamical Friction in a Fuzzy Dark Matter Universe**
Lancaster, L.; Giovanetti, C.; Mocz, P.; Kahn, Y.; Lisanti, M.; Spergel, D.N.; 2020 JCAP, 01, 001

28. **First star-forming structures in fuzzy cosmic filaments**
Mocz, P.; Fialkov, A.; Vogelsberger, M.; Becerra, F.; Amin, M.A.; Bose, S.; Boylan-Kolchin, M.; Chavanis, P.-H.; Hernquist, L.; Lancaster, L.; Marinacci, F.; Robles, V.H.; Zavala, J.; 2019 Phys. Rev. Lett. **(Editors' Suggestion)**, 123, 141301

27. **A Markov model for non-lognormal density distributions in compressive isothermal turbulence**
Mocz, P.; Burkhart, B.; 2019 ApJL, 884, L35

26. **Formation, Gravitational Clustering and Interactions of Non-relativistic Solitons in an Expanding Universe**
Amin, M.A.; Mocz, P.; 2019 Phys. Rev. D, 100, 063507

25. **The Self-gravitating Gas Fraction and The Critical Density for Star Formation**
Burkhart, B.; Mocz, P.; 2019 ApJ, 879, 129

24. **Heating of Milky Way disc Stars by Dark Matter Fluctuations in Cold Dark Matter and Fuzzy Dark Matter Paradigms**
Church, B.V.; Mocz, P.; Ostriker, J.P.; 2019 MNRAS, 485, 2861

23. **Star formation from dense shocked regions in supersonic isothermal magneto-turbulence**
Mocz, P.; Burkhart, B.; 2018 MNRAS, 480, 3916

22. **Evolution of the Black Hole Mass Function in Star Clusters from Multiple Mergers**
Christian, P.; Mocz, P.; Loeb, A.; 2018 ApJL, 858, L8

21. **Non-ideal magnetohydrodynamics on a moving mesh**
Marinacci, F.; Vogelsberger, M.; Kannan, R.; Mocz, P.; Pakmor, R.; Springel, V.; 2018 MNRAS, 476, 2476

20. **Schrödinger-Poisson–Vlasov-Poisson correspondence**
Mocz, P.; Lancaster, L.; Fialkov, A.; Becerra, F.; Chavanis, P.-H.; 2018 Phys. Rev. D, 97, 083519

19. **Galaxy Formation with BECDM - I. Turbulence and relaxation of idealised haloes**
Mocz, P.; Vogelsberger, M.; Robles, V.H.; Zavala, J.; Boylan-Kolchin, M.; Fialkov, A.; Hernquist, L.; 2017 MNRAS, 471, 4559

18. **Unveiling the role of the magnetic field at the smallest scales of star formation**
Hull, C.L.H.; Mocz, P.; Burkhart, B.; Goodman, A.A.; Girart, J.M.; Cortés, P.C.; Hernquist, L.; Springel, V.; Li, Z.-Y.; Lai, S.-P.; 2017 ApJL, 842, L9

17. **Moving mesh simulations of star forming cores in magneto-gravo-turbulence**
Mocz, P.; Burkhart, B.; Hernquist, L.; McKee, C.F.; Springel, V.; 2017 ApJ, 838, 40

16. **Integer lattice dynamics for Vlasov-Poisson**
Mocz, P.; Succi, S.; 2017 MNRAS, 465, 3154

15. **Correspondence between constrained transport and vector potential methods for MHD**
Mocz, P.; 2017 J. Comp. Phys., 328, 221

14. **A moving mesh unstaggered constrained transport scheme for MHD**
Mocz, P.; Pakmor, R.; Springel, V.; Vogelsberger, M.; Marinacci, F.; Hernquist, L.; 2016 MNRAS, 463, 477

13. **Improving the convergence properties of the moving-mesh code AREPO**
Pakmor, R.; Springel, V.; Bauer, A.; Mocz, P.; Munoz, D.J.; Ohlmann, S.T.; Schaal, K.; Zhu, C.; 2016 MNRAS, 455, 1134

12. **The large-scale properties of simulated cosmological magnetic fields**
Marinacci, F.; Vogelsberger, M.; Mocz, P.; Pakmor, R.; 2015 MNRAS, 453, 3999

11. **Reducing noise in moving-grid codes with strongly-centroidal Lloyd mesh regularization**
Mocz, P.; Vogelsberger, M.; Pakmor, R.; Genel, S.; Springel, V.; Hernquist, L.; 2015 MNRAS, 452, 3853

10. **Numerical solution to the non-linear Schrödinger equation using smoothed-particle hydrodynamics**
Mocz, P.; Succi, S.; 2015 Phys. Rev. E, 91, 053304

9. **Interpreting MAD within multiple accretion regimes**
Mocz, P.; Guo, X.; 2015 MNRAS, 447, 1498

8. **A constrained transport scheme for MHD on unstructured static and moving meshes**
Mocz, P.; Vogelsberger, M.; Hernquist, L.; 2014 MNRAS, 442, 43

7. **Do high-redshift quasars have powerful jets?**
Fabian, A.C.; Walker, S.A.; Celotti, A.; Ghisellini, G.; Mocz, P.; Blundell, K.M.; McMahon, R.G.; 2014 MNRAS, 442, L81

6. **A discontinuous Galerkin method for solving the fluid and magnetohydrodynamic equations in astrophysical simulations**
Mocz, P.; Vogelsberger, M.; Sijacki, D.; Pakmor, R.; Hernquist, L.; 2014 MNRAS, 437, 397

5. **Cosmological growth and feedback from supermassive black holes**
Mocz, P.; Fabian, A.C.; Blundell, K.M.; 2013 MNRAS, 432, 3381

4. **The Tully-Fisher relation for 25,000 Sloan Digital Sky Survey galaxies as a function of environment**
Mocz, P.; Green, A.; Malacari, M.; Glazebrook, K.; 2012 MNRAS, 425, 296

3. **The inverse-Compton ghost HDF 130 and the giant radio galaxy 6C 0905+3955: matching an analytic model for double-lobed radio source evolution**
Mocz, P.; Fabian, A.C.; Blundell, K.M.; Goodall, P.T.; Chapman, S.C.; Saikia, D.J.; 2011 MNRAS, 417, 1576

2. **Inverse-Compton ghosts and double-lobed radio sources in the X-ray sky**
Mocz, P.; Fabian, A.C.; Blundell, K.M.; 2011 MNRAS, 413, 1107

1. **A Detection of an X-ray Wind and an Ionized Disk in the Chandra HETGS Observation of the Seyfert 2 Galaxy IRAS 18325-5926**
Mocz, P.; Lee, J.C.; Iwasawa, K.; Canizares, C.R.; 2011 ApJ, 729, 30

Grants and Proposals

2022 **PI of LEARN grant**, *"High-Order ALE Compressible Magnetohydrodynamics"*, LLNL.
(est. value $120,000)

2021 **Co-PI of DiRAC computing grant**, *"Quantifying the effects of fuzzy dark matter on the formation of first galaxies"*, ACSP219.
5.4M cpu core-hrs on *Cumulus* Data Intensive@Cambridge (est. value $30,000). PI: Anastasia Fialkov

2021 **Co-PI of NSF grant**, *"Collaborative Research: Constraining Fuzzy Dark Matter with Cosmological Simulations"*, NSF 18-575.
With UT Austin and MIT. (est. value $500,000). PI: Mike Boylan-Kolchin

2020 **Co-PI of DiRAC computing grant**, *"Cosmological Simulation of Fuzzy Dark Matter with Self-Interaction"*, ACSP219.
4M cpu core-hrs on *Cumulus* Data Intensive@Cambridge (est. value $28,000). PI: Anastasia Fialkov

2017 **Co-PI of TACC *Stampede2* computing grant**, *"Cosmology with Bose-Einstein Condensate Dark Matter: Cosmic Web and First Galaxies"*, TG-AST170020.
60,750 node-hours ($\sim$ 3M cpu core-hrs; est. value $18,000). PI: Anastasia Fialkov

2017 **PI of TACC *Stampede2* computing grant**, *"The Formation of First Stars in Bose-Einstein Condensate Dark Matter"*, TG-AST170015.
50,000 CPU core-hrs (est. value $1,800)

2017–2020 **PI of NASA Einstein Fellowship award**, *"Moving mesh magnetohydrodynamics: understanding the role of magneto-turbulence in cosmological structure formation and star formation"*, PF7-180164.
Total budget: $300,000. Host: Jim Stone

2015 **Co-PI of Gauss Centre for Supercomputing grant**, *"Predicting galaxy formation in a representative volume of the Universe"*.
92M cpu core-hrs on *Hornet* in Stuttgart. PI: Volker Springel

2015–2017 **Co-PI of NASA NESSF award**, *"Moving Mesh Cosmology with Magnetohydrodynamics"*, NNX15AR88H.
Total budget: $100,000. Admin PI: Lars Hernquist

2012–2015 **PI of NSF GRFP award**, *"Jet dynamics and kinetic feedback from supermassive black holes"*, DGE-1144152.
Total budget: $100,000.

Presentations

Invited Talks

11/2025 **Axions in Japan**, *IPMU*, "Axion Structure Formation: Self-Interactions, Multi-Field Dynamics, and Observational Signatures".

03/2025 **CTC Seminar**, *University of Maryland*, "Differentiable Astrophysics Simulations".

04/2024 **Applied Math Seminar**, *Harvard*, "Multi-physics simulations in the era of high-performance computing and AI".

01/2023 **PPPL Seminar**, *PPPL*, "Moving Mesh Magnetohydrodynamics and other Numerical Methods for Astrophysics and Plasmas".

03/2022 **Astronomy Colloquium**, *ANU*, "Fuzzy Dark Matter Cosmology".

02/2022 **LEPP Seminar**, *Cornell*, "Next-Generation Multi-Physics Simulation Codes for Computational Astrophysics ".

04/2021 **Eisenstein Group Meeting**, *Harvard*, "Cosmological Simulations with Quantum Computers".

03/2021 **Carnegie Mellon University Colloquium**, *Pittsburgh*, "Next-Generation Astrophysical Simulation Methods".

03/2021 **NYU X CCA cosmology X data science**, *Flatiron Institute*, "Cosmological Simulations with Quantum Computers".

03/2021 **University of Patras Physics Seminar**, *Patras, Greece*, "First Galaxies in Cold, Warm, and Fuzzy Dark Matter Cosmologies".

03/2021 **University of Florida Colloquium**, *Gainesville*, "Next-Generation Astrophysical Simulation Methods".

02/2021 **Thunch Talk**, *Princeton*, "Cosmological Simulations with Quantum Computers".

12/2020 **Los Alamos National Lab Seminar**, *Livermore*, "Moving Mesh Magnetohydrodynamics and other methods for Computational Astrophysics".

12/2020 **Lawrence Livermore National Lab WCI Seminar**, *Livermore*, "Moving Mesh Magnetohydrodynamics and other methods for Computational Astrophysics".

11/2020 **Xanadu Seminar**, *Toronto*, "Towards Cosmological Simulations of Dark Matter on Quantum Computers".

11/2020 **The Chinese University of Hong Kong Colloquium**, *Hong Kong*, "First Galaxies in Cold, Warm, and Fuzzy Dark Matter Cosmologies".

09/2020 **Lawrence Livermore National Lab Astro Seminar**, *Livermore*, "First Structures in Cold, Warm, and Fuzzy Dark Matter".

07/2020 **Fuzzy Dark Matter Workshop**, *Göttingen*, "First Structures in Fuzzy Dark Matter".

05/2020 **Physics and Math Seminar**, *Duke University*, "Fuzzy Dark Matter Cosmology".

04/2020 **University of Florida Colloquium**, *Gainesville*, "Fuzzy Dark Matter Cosmology".

02/2020 **Gravity Group Seminar**, *Princeton*, "First Galaxies in Fuzzy Dark Matter".

02/2020 **Magnetic Fields in the Universe 7**, *Vietnam*, "Studying Dense Structures in a Turbulent Interstellar Medium with a Moving Mesh".

02/2020 **Cornell Astrophysics Colloquium**, *Cornell*, "Is dark matter cold, warm, or fuzzy?".

02/2020 **UCSC Astrophysics Seminar**, *Santa Cruz*, "Fuzzy Dark Matter Cosmology".

01/2020 **CCA Astrophysics Seminar**, *Flatiron Institute*, "The Art of Scientific Computing (for Astrophysics)".

01/2020 **NYU Astrophysics Seminar**, *New York University*, "Fuzzy Dark Matter Cosmology".

01/2020 **McWilliams Astro-Seminar**, *Carnegie Mellon University*, "How first galaxies can reveal whether dark matter cold, warm, or fuzzy".

09/2019 **Competing Structure Formation Models Workshop**, *Reykjavik*, "Fuzzy Dark Matter: overview".

09/2019 **Rutgers Astronomy Seminar**, *Rutgers*, "Ultralight Dark Matter".

08/2019 **Cosmo Cruise**, *Italy/Greece*, "Ultralight Dark Matter".

02/2019 **UW-Madison Astrophysics Colloquium**, *Madison*, "Structure formation and turbulent processes in the Universe".

01/2019 **Big Apple Magnetic Fields Conference**, *CCA Flatiron*, "Shock structures in magnetized supersonic isothermal turbulence".

12/2018 **Columbia Physics Theory Seminar**, *Columbia*, "Small-scale features in fuzzy dark matter".

11/2018 **MX Dark Matter Conference**, *Cancun, Mexico*, "Quantum Wave Dark Matter and the Classical Limit".

10/2018 **Rice Theory Seminar**, *Rice University*, "Small-scale structure in fuzzy dark matter and the classical limit".

09/2018 **CITA Theory Seminar**, *Toronto*, "Small-scale structure in fuzzy dark matter and the classical limit".

03/2018 **Ringberg Computational Galaxy Formation**, *Germany*, "Galaxy Formation with Bose-Einstein Condensate Dark Matter".

09/2017 **Brown BASS Talk**, *Brown*, "Galaxy Formation with Axion Dark Matter".

12/2016 **Cosmic Rays, Astrophysical Turbulence and Magnetic Reconnection Conference, IIP**, *Natal, Brazil*, "Moving mesh simulations of star forming cores in magneto-gravo-turbulence".

01/2015 **Istituto per le Applicazioni del Calcolo "Mauro Picone" Theory Talk**, *Rome*, "Moving Mesh and Smoothed Particle Methods for Computational Fluid Dynamics".

Contributed Talks

06/2025 **AAS Meeting**, *Anchorage*, "Differentiable Astrophysics".

11/2024 **Supercomputing**, *Atlanta*, "Multi-physics simulations with JAX".

11/2023 **APS DPP**, *Denver*, "High-order finite element arbitrary Lagrangian-Eulerian resistive magnetohydrodynamics coupled to a multi-physics code".

10/2020 **Galaxy Formation and Evolution in the Era of the Nancy Grace Roman Space Telescope**, *Maryland*, "First Galaxies as Probes of Dark Matter Physics".

01/2020 **AAS Meeting**, *Honolulu*, "First star-forming structures in fuzzy dark matter".

12/2019 **Turbulence Across Vast Scales**, *CCA Flatiron*, "Dense structures in compressible turbulence".

11/2019 **Cosmic Turbulence and Magnetic Fields**, *Cargese*, "Pre-stellar core formation from dense shocked regions in supersonic isothermal magnetoturbulence".

11/2019 **Einstein Symposium**, *Washington DC*, "First star-forming structures in fuzzy dark matter cosmic filaments".

10/2019 **Cosmic Controversies**, *Chicago KICP*, "First star-forming structures in fuzzy cosmic filaments".

08/2019 **UCSC Galaxy Workshop**, *Santa Cruz*, "First structures in ultralight dark matter".

06/2019 **UH Physics Colloquium**, *University of Hawaii*, "First star-forming structures in fuzzy cosmic filaments".

10/2018 **Einstein Symposium**, *Harvard*, "Small-scale features in fuzzy dark matter cosmology".

02/2018 **ALMA NA-Taiwan Joint Workshop: Magnetic Fields or Turbulence?**, *Taiwan*, "Magneto- and turbulent regimes of star formation".

10/2017 **CIRM Collisionless Boltzmann (Vlasov) Equation and Modeling of Self-Gravitating Systems and Plasmas**, *Marseilles*, "Solving Vlasov-Poisson dynamics on an integer lattice".

10/2017 **Einstein Symposium**, *Harvard*, "The role of magneto-turbulence in star formation".

09/2017 **CCA NY Area Computational Hydro Workshop**, *CCA Flatiron*, "Galaxy Formation with Axion Dark Matter".

05/2017 **Harvard ITC Luncheon**, *Harvard*, "Integer Lattice for Vlasov-Poisson".

05/2017 **Harvard ITC Luncheon**, *Harvard*, "Quantum Turbulence in Bose-Einstein Condensate Dark Matter".

11/2016 **Berkeley TAC Seminar**, "Moving mesh simulations of star forming cores in magneto-gravo-turbulence".

09/2016 **Harvard ITC Luncheon**, "Moving mesh simulations of star forming cores in magneto-gravo-turbulence".

06/2016 **Astronum**, *Monterey, CA*, "Moving mesh magnetohydrodynamics".

05/2016 **Crutcher & Heiles Conference**, *Madison*, "Moving mesh magnetohydrodynamics and applications to star forming cores".

06/2013 **Southern Cross Conference Series VI: Feeding, Feedback, and Fireworks: Celebrating Our Cosmic Landscape**, *Hamilton Island, Australia*, "A discontinuous Galerkin method for solving the fluid and MHD equations in astrophysical simulations".

08/2011 **Swinburne CAS Seminar**, *Swinburne Univ. of Technology*, "Tully-Fisher Relationships for SDSS Galaxies as a Function of Environment".

07/2011 **IoA X-Ray Group Talk**, *University of Cambridge*, "Cosmological growth and feedback of massive black holes".

05/2011 **Research Presentation**, *Harvard*, "SMA Observation of the Extended Emission in the High-Mass Star Forming Region AFGL 2591".

04/2011 **Junior Thesis Presentations**, *Harvard*, "X-ray spectroscopy of silicate dust in the ISM and environments around XRBs".

03/2011 **Research Presentation**, *Harvard*, "Laboratory and Astronomical Observations of the CN Radical".

08/2010 **IoA X-Ray Group Talk**, *University of Cambridge*, "Double radio sources and inverse-Compton ghosts in the X-ray sky".

08/2009 **PRISE Talk**, *Harvard*, "A Search for X-Ray Winds and Strong Gravity Around a Supermassive Black Hole In A Distant Galaxy".

Whitepapers/LOIs

2021 **Snowmass**, "Cosmic probes of ultra-light axion dark matter".

2021 **Snowmass**, "A simulation program to discover dark matter physics in the sky".

Professional Skills

Codes Developer: MARBL, AREPO, MESA

Web HTML5, CSS

Programming Languages C++, C, Python, JAX, Lua, Fortran, Matlab, Mathematica, Javascript

Paradigms MPI, OpenMP, CUDA, OpenACC, Kokkos/RAJA

High-performance computing Use of several large supercomputing facilities, including TACC *Stampede2*, *DiRAC*, *SuperMUC*, utilizing 10,000s cores and 10s millions of cpu-core hours. Led successful supercomputing proposals.

Certificates NVIDIA Deep Learning Institute

- Fundamentals of Accelerated Computing with Modern CUDA C++
- Fundamentals of Deep Learning
- Data Parallelism: How to Train Deep Learning Models on Multiple GPUs

Teaching

Spring 2016 **Astronomy 151. Astronomical Fluid Dynamics**, *Teaching Fellow*, Harvard.

- student evaluation score 5.0/5.0, 5 students
- duties: office-hours, special topics sections, grading, homework solutions

Fall 2014 **Applied Computation 274. Computational Fluid Dynamics**, *Section leader*, Harvard.

- student evaluation score 4.5/5.0, 6 students
- duties: lectures, office-hours, course material and homework development, grading, final project supervision

Spring 2014 **Applied Computation 274. Computational Fluid Dynamics**, *Section leader*, Harvard.

- student evaluation score 4.0/5.0, 6 students
- duties: office-hours, course material and homework development, grading, final project supervision

Fall 2012 **Applied Mathematics 205. Advanced Scientific Computing: Numerical Methods**, *Section leader*, Harvard.

- student evaluation score 4.7/5.0, 56 students
- duties: weekly section, office-hours, course material and homework development, final project supervision

Mentoring

2023 **Eric Ludwig**, *graduate*, RPI, Astrophysics.

2023 **Seth Webb**, *undergraduate*, Texas A& M, Physics.

2022 **William White**, *graduate*, University of Michigan, Applied Mathematics.

2021– **Connor Painter**, *graduate*, UT Austin, Astrophysics.

2021– **Tibor Dome**, *graduate*, Cambridge, Astrophysics.

2021–2022 **Rohith Karur**, *undergraduate*, UC Berkeley, Physics.

2020– **Hayden Foote**, *graduate*, University of Arizona, Astrophysics.

2020–2021 **Noah Notis**, *undergraduate*, Princeton, Physics.

2020–2021 **Benjamin Hamm**, *graduate*, Duke, Physics.

2020–2022 **James Beattie**, *graduate*, ANU, Astrophysics.

2019– **Michael Foley**, *graduate*, Harvard, Astrophysics.

2019–2020 **Elliot Davies**, *undergraduate*, Princeton, Physics.

2019 **Cara Giovanetti**, *undergraduate*, Princeton, Physics.

2018 **Ben Church**, *undergraduate*, Columbia, Mathematics & Physics.

2017–2019 **Lachlan Lancaster**, *graduate*, Princeton, Astrophysics.

2016 **Alex Gurvich**, *undergraduate*, Harvard, Physics & Astrophysics.

2016 **Sruthi Narayanan**, *undergraduate*, MIT, Computer Science.

Service and Outreach

2022–2023 **LLNL DSTI Research Program Mentor** Advised graduate student research project in scientific computing.

2021–2022 **NASA Cosmic Origins Transitional Leadership Team.** Stars Science Interest Group for decadal survey.

2020– **Medium Articles.** Published introductory guides (undergraduate level) on scientific computing and simulation methods.

2020 **Skype A Scientist.** Connected virtually with teachers and classrooms to answer student questions in astronomy.

2020–2021 **Postdoc Mentor.** Provided professional mentorship for incoming postdocs at Peyton and helped design postdoc handbook.

2019–2020 **Princeton Astrophysics Undergraduate Summer Research Program Mentor.** Advised physics students on summer research project on scalar field dark matter, leading to publication.

2019–2021 **Princeton Astro-ph Coffee Discussion Leader.**

2019 **Press Release.** Joint Research Press Release with Princeton, MIT, Cambridge, UT Austin, Nature regarding first simulations full-physics simulations of fuzzy dark matter.

2019 **Guest Lecturer.** Taught graduate students about moving mesh computational fluid dynamics with interactive coding demos in Princeton graduate seminar course AST 542.

2017–2020 **Princeton Star-formation and ISM Rendezvous Seminar Organizer.**

2015–2017 **Harvard Astronomy Department Peer Mentor.** Served as a guide and resource for first and second year graduate student mentees.

2015–2017 **Harvard-Smithsonian CfA Library Committee Representative.** Helped advise head librarian and committee of 8 scientists on the future of the library, new digital services, journal subscriptions, educational services, and resources for graduate students.

2015 **Einstein in the Classroom Instructor.** Engaged with Pierce Middle School in the greater Boston area to offer physics activities in the classroom covering relativity, space-time curvature, the life cycles of stars, and the relative sizes of the objects in the observable Universe.

2014– **Peer Review.** Physical Review Journals (PRD, PRL), Astrophysical Journal, Monthly Notices of the Royal Astronomical Society, Journal of Open Source Software, Scientific Reports, Journal of Computational Physics, Complex Systems

2014– **Proposal Referee/Panelist.** NASA Earth and Space Science Fellowship (NESSF), Future Investigators in NASA Earth and Space Science and Technology (FINESST), NASA ADAP/ROSES, French National Research Agency (ANR)

Memberships

2024– **Association for Computing Machinery**.

2020– **American Physical Society**.

2011– **Phi Beta Kappa**.

2006– **American Astronomical Society**.